

Practical Power Design for Embedded & Robotics Systems

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First Edition

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Copyright & Revision History

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Copyright Notice

This handbook was created for educational and project development purposes. It is intended to provide practical guidance on power design concepts commonly encountered in embedded systems, robotics, and electronics projects.

The information contained in this handbook is based on practical engineering principles and real-world applications. While every effort has been made to ensure accuracy, readers are encouraged to verify specifications, calculations, and safety requirements for their specific applications.

Preface

Power-related issues are among the most common causes of failure in embedded systems and robotics projects. Many seemingly complex problems (random resets, unstable sensors, communication errors, overheating components, and unreliable motors) can often be traced back to fundamental power design mistakes.

This handbook was created to provide a practical introduction to power electronics concepts from the perspective of a project builder. Rather than focusing on semiconductor physics, advanced converter design, or mathematical derivations, the emphasis is placed on understanding how power-related components behave in real systems and how they should be applied to create reliable designs.

The material presented here is intended for students, hobbyists, robotics teams, makers, and embedded systems developers who wish to build systems that work not only in theory, but also in practice.

The goal of this handbook is simple:

To help readers design electronics that are reliable, efficient, and robust enough for real-world operation.

How To Use This Handbook

This handbook is structured as a practical reference rather than a traditional academic textbook. Each chapter focuses on a specific aspect of power design commonly encountered in embedded and robotics projects.

Recommended Reading Order

Readers new to the subject are encouraged to read the chapters sequentially:

1. Capacitors
2. Inductors
3. Voltage Regulators
4. Diodes
5. MOSFETs
6. Grounding & Power Distribution
7. Noise, EMI & Filtering
8. Battery & Protection Design
9. PCB Power Layout Rules
10. Reliability & Thermal Considerations

These chapters establish the core concepts required to understand practical power system design.

Using the Checklist Chapter

Chapter 11, System-Level Power Design Checklist, is intended to be used during project development and troubleshooting.

It can serve as:

- A pre-power-up inspection checklist
- A design review checklist
- A troubleshooting guide
- A reliability verification tool

Scope of This Handbook

This handbook focuses on practical application.

Topics such as:

- Semiconductor physics
- Control theory
- Converter compensation
- Advanced power supply design

are intentionally outside its scope.

The objective is not to teach power electronics research or converter design, but to provide the knowledge required to successfully apply power electronics concepts in real projects.

Design Philosophy

The following principles guided the creation of this handbook and should guide the design of reliable electronic systems.

Principle 1: Reliability Over Complexity

A simple system that operates reliably is often preferable to a more complex system that offers additional features but introduces unnecessary failure points.

Reliability should always be considered a primary design objective.

Principle 2: Protect Before Powering

Protection circuits are significantly easier and less expensive to implement before a failure occurs than after one.

Whenever practical, systems should include protection against:

- Reverse polarity
- Overcurrent conditions
- Voltage spikes
- Battery misuse

Principle 3: Design For Worst-Case Conditions

A circuit should not only operate under ideal laboratory conditions.

It should continue to function correctly when:

- Batteries are partially discharged
- Loads suddenly increase
- Ambient temperature rises
- Electrical noise is present

Reliable systems are designed for worst-case scenarios, not best-case scenarios.

Principle 4: Leave Engineering Margin

Components should not be operated continuously at their maximum ratings.

Adequate margin should be maintained for:

- Voltage
- Current
- Power dissipation
- Temperature

Margin improves both safety and reliability.

Principle 5: Verify Assumptions With Measurements

Assumptions should be confirmed through testing whenever possible.

Measurements are often more valuable than predictions.

Useful measurements include:

- Supply voltage
- Current consumption
- Temperature
- Battery performance

Engineering decisions should be based on evidence rather than assumptions.

Principle 6: Simplicity Is A Feature

Every additional component, wire, connector, or subsystem introduces complexity.

Good engineering seeks the simplest solution capable of meeting the requirements of the problem.

Simple systems are generally:

- Easier to understand
- Easier to troubleshoot
- Easier to maintain
- More reliable

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Chapter One: Capacitors

1. Why Are Capacitors Needed?

In real systems (motors, servos, microcontrollers):

- Loads draw non-constant current
- Motors create sudden spikes
- Regulators are not ideal
- Wires have resistance and inductance

This causes:

- Voltage dips
- Noise
- Brownouts
- EMI issues

Capacitors are added to:

- Supply short bursts of energy
- Reduce voltage ripple
- Suppress high-frequency noise

2. Fundamental Equation

$$I = C \frac{dV}{dt} \Rightarrow C = \frac{I \cdot \Delta t}{\Delta V}$$

Where:

- I = current spike
- Δt = duration of spike
- ΔV = maximum allowed voltage drop
- C = required capacitance

Interpretation

- Larger current $\rightarrow$ larger capacitor
- Longer spike $\rightarrow$ larger capacitor
- Smaller allowed voltage drop $\rightarrow$ larger capacitor

3. Types of Capacitors

A. Bulk Capacitors (Electrolytic):

- a. Large (100 μF – several 1000 μF)
- b. Handle millisecond scale current spikes
- c. Used near:
 - i. Motor drivers
 - ii. Servo rails
 - iii. Regulator inputs
- d. Purpose: Reduce voltage sag during sudden load changes.

B. Decoupling Capacitors (Ceramic):

- a. Small ($\sim 0.1\mu\text{F}$)
- b. Very low ESR and ESL
- c. Placed close to IC/logic power pins
- d. Purpose
 - i. Remove high frequency switching noise
 - ii. Stabilize digital circuits

4. Placement Principles

Rule 1: Close to the Load

Capacitors must be placed:

- Physically close to the device drawing current
- With short traces

Otherwise wire inductance reduces effectiveness.

Rule 2: Use Both Bulk and Ceramic

Typical good practice:

- 470-1000 μF Electrolytic
- 0.1 μF ceramic in parallel

Electrolytic -> stores energy

Ceramic -> filters high frequency noise

Rule 3: Distribute Capacitance

Use multiple smaller capacitors near each module rather than one big large capacitor near the battery.

Distributed Capacitance reduces local voltage drops.

Chapter Two: Inductors

1. Why are Inductors Needed?

In real systems:

- Buck Converters switch ON/OFF rapidly
- Motors are inductive loads
- Current changes rapidly in switching circuits

Rapid Change in Current causes the following:

- Voltage spikes
- Switching instability
- EMI radiation
- Ripple current

Inductors do the following:

- Resist sudden change in current
- Store energy in magnetic field
- Smooth current in switching converters
- Filter ripple in LC filters

2. Fundamental Equation

$$V = L \frac{dI}{dt} \Rightarrow \frac{dI}{dt} = \frac{V}{L}$$

Where:

V = Voltage across Inductor

L = Inductance

$\frac{dI}{dt}$ = rate of change of current

Interpretation

Higher inductance → slower current change

Higher applied voltage → faster current change

Inductors resist current change

This is why:

- Buck converters use inductors to smooth current
- Motors generate back EMF when current changes

3. Energy Storage

$$E = \frac{1}{2}LI^2$$

Inductors store energy in the form of magnetic fields.

When the switch is turned off, energy must go somewhere -> diode conducts current.

4. Key Parameters

a. Inductance:

Units: Henry

b. Saturation Current:

If current is more than saturation rating:

- Inductance collapses
- Current spikes
- Converter fails

Rule to remember: always maintain saturation current > peak current

c. DC Resistance (DCR):

Internal resistance of the inductor.

Causes power loss:

$$P = I^2R$$

Where R = DCR of inductor

For higher efficiency, DCR should be lower.

d. Core Type:

- Ferrite - for higher frequency and higher efficiency power applications
- Powdered iron - for higher saturation current applications, like switching power supplies.
- Shielded vs. unshielded inductors - shielded inductors produce less EMI

5. Placement Principles

Rule 1: Keep switching loop small

Rule 2: Place near MOSFET and diode

Rule 3: Avoid routing signal traces under the inductor

Common Mistakes:

- Choosing inductor based only on inductance
- Ignoring saturation current
- Using large unshielded inductor near MCU
- Ignoring DCR losses

Chapter Three: Voltage Regulators

1. Why are Voltage Regulators Needed?

In real systems:

- Batteries discharge over time
- Motors cause sudden voltage dips
- Adapters contain ripple and noise
- Long wires introduce resistance and inductance

And Microcontrollers, sensors, RF modules, ADCs and logic ICs require stable, precise voltage.

Without regulation:

- Brownouts
- Random resets
- Communication failure
- ADC inaccuracy
- Permanent damage

Voltage regulators ensure constant output voltage regardless of varying inputs, loads, and temperature.

2. Types of Voltage Regulators

- Linear Regulators
- Switching Regulators
 - Buck (Step-down)
 - Boost (Step-up)
 - Buck-Boost

3. Linear Regulators:

a. Why are Linear Regulators used?

They are simple, low noise, and ideal for small loads.

Common examples of linear regulators are 7805, AMS1117, and LM317.

They are used in sensor rails, analog circuits, and low current logic supplies.

b. How do they work?

They act like a controlled variable resistor. And drop excess voltage as heat

$$V_{drop} = V_{in} - V_{out}$$

c. Power dissipation

$$P_{lost} = (V_{in} - V_{out}) \cdot I_{load}$$

This is the most important formula that governs the working principle of linear regulators.

For example, if the input voltage is $12V$ and required output voltage is $5V$ with a current of $1A$ flowing through the circuit.

$$P = (12 - 5) \cdot 1 = 7W$$

I.e., $7W$ power must be dissipated as heat. Hence the reason small regulators heat up quickly.

d. Efficiency

$$\eta = \frac{V_{out}}{V_{in}}$$

Considering the same example as above, we get

$$\eta = 41\%$$

Meaning, 59% is wasted as heat.

e. Dropout voltage

The minimum difference between required output voltage and input voltage required for a linear regulator to be used correctly is called dropout voltage, often given by:

$$V_{dropout} \leq V_{out} - V_{in}$$

For example, the dropout voltage for AMS1117 is around $1.1V$

Hence, to get a $5V$ output, the input must be at least $6.1V$

If the input falls below this value, the output voltage begins to drop and regulation is lost.

f. When to use linear regulators

Linear regulators should be used when the following conditions are met:

- Current $< 300mA$
- Input voltage is close to output voltage
- Low noise is required
- Heat is manageable

In case of any of the following conditions, it is suggested to avoid the use of linear regulators:

- High current ($> 500mA$)
- Large voltage drop is required
- Battery powered efficiency matters more

4. Switching Regulators (Buck Converters):

a. Why are switching regulators needed?

In robotics we have battery powered systems, high current motors and systems where efficiency is critical and heat must be minimized. Due to these requirements, the use of linear regulators is not advised.

In such situations, switching regulators, more commonly known as Buck Converters, are used as they can provide 85-95% efficiency.

b. Buck converter (step down)

A buck converter reduces output voltage efficiently when compared to a linear regulator.

Example:

24V -> 5V

12V -> 3.3V

c. Fundamental Relationship

The fundamental formula connecting the input and output voltage of a buck converter is:

$$V_{out} = D \cdot V_{in}$$

Where D is called duty cycle and ranges from 0 to 1.

d. How it works

A buck converter rapidly switches the input voltage on and off.

An inductor and capacitor smooth this pulsed energy into a lower DC voltage.

By adjusting the percentage of time the switch remains ON (duty cycle), the converter controls the output voltage.

e. Input & Output Capacitors

Buck converters draw current in short pulses.

Input capacitors provide these current bursts locally and reduce the stress on the supply wiring.

Output capacitors smooth switching ripple and help the regulator respond to sudden load changes.

Practical rule:

- Place capacitors close to the regulator

- Follow datasheet recommendations
- Never omit required capacitors

f. When to use buck regulators?

A buck regulator is used when the following conditions is satisfied:

- Current > 300-500 mA
- Large voltage drop is required
- Battery powered system
- Efficiency matters

5. Linear vs Buck decision rule

Small current + low noise -> use linear regulator

High current + efficiency required -> use buck converter

Chapter Four: Diodes

1. Why are Diodes Needed?

In electronic systems, current often needs to flow in only one direction. Without control of current directions batteries can be connected incorrectly, motors can generate damaging voltage spikes, and sensitive components can be damaged.

Diodes are used to make current flow unidirectional, protect circuits from reverse polarity and voltage spikes, and convert AC to DC.

2. What is a Diode?

A diode is a semiconductor device that acts like a one way valve for current. Current flows from anode to cathode in a diode, and the current is blocked in the opposite direction.

3. Forward Voltage Drop

Since real diodes are, well, real and not ideal. When current flows through it, a small voltage is lost.

Following are typical values for forward voltage loss:

Diode Type	Forward Voltage
Silicon Diode	~0.7V
Schottky Diode	~0.2V-0.4V
LED	Depends on color

Example: A 5v supply through a silicon diode will give approximately 4.3V output instead of a 5V output in the forward bias mode.

4. Types of Diodes Used in Practical Systems

General Purpose Diodes

Common general purpose diodes are 1N4001 and 1N4007. They are used for reverse polarity protection, and basic rectification.

Although these diodes are cheap and easily available, they pose a slow switching speed.

Schottky Diodes

Schottky does have a lower forward voltage drop and faster switching compared to general purpose silicon diodes.

They are primarily used in buck converters, battery powered systems, and high efficiency designs.

Since the forward voltage drop is lower, schottky diodes generate less heat and cause less power loss.

LEDs (Light Emitting Diodes)

LEDs emit light when current flows through them.

They are used for indicators, displays, and lighting.

Note: LEDs always require current limiting.

This is usually done using a resistor or a constant current driver.

5. Reverse Polarity Protection

One of the most common reasons for your circuit to get damaged is incorrect battery connection.

To prevent this, a diode can be used.

When battery polarity is correct, the current flows normally; when battery polarity is reversed, current is blocked and no components are damaged.

6. Flyback Diodes

Why Are Flyback Diodes Needed?

Motors, relays and solenoids are inductive loads, they resist sudden changes in current. When current is suddenly interrupted, a large voltage spike is generated by these loads. This spike can damage MOSFETs and MCUs, and cause noise and resets.

How Flyback Diodes work

A diode is placed across the inductive load.

During normal operation, the diode does nothing; when power is removed, the stored energy circulates through the diode instead of creating a large voltage spike.

This protects the switching device.

Practical Rule

Whenever switching DC motors, relays, solenoids: always check whether flyback protection is present.

Many driver modules include it internally.

7. Rectification (Brief Overview)

Diodes can also be used to convert AC into DC.

These rectification circuits are present in power adapters, power supplies, and battery chargers.

8. Choosing a Diode

When selecting a diode for a particular circuit, we must check the following:

Current Rating

Must exceed expected current.

Reverse Voltage Rating

Must exceed the maximum voltage the diode will block.

Forward Voltage Drop

Lower drop leads to higher efficiency and less heat generated.

Switching Speed

Switching speed considerations are important for switching regulators and high frequency circuits.

Chapter Five: MOSFETS

1. Why are MOSFETs needed?

MCUs can control signals, and cannot directly power large loads.

For example:

ESP32 GPIO runs at 3.3V and recommended 12mA.

While motors run at 6V-24V and require hundreds of mA to sometimes several A.

Hence, direct connection is impossible.

A MOSFET allows a small control signal to switch a much larger load.

Therefore, MOSFETs are used to control motors, switch LEDs, drive solenoids, control relays, build motor drivers and switching regulators.

2. What is a MOSFET?

MOSFET stands for:

Metal Oxide Semiconductor Field Effect Transistor

It can be thought of as an electronically controlled switch.

3. MOSFET Terminals

Gate (G)

Gate is the control terminal responsible for deciding whether the switch is to be turned on or off.

Very little current flows into the gate.

Drain (D)

This is the terminal through which the current enters.

Source (S)

This is the terminal through which the current exits.

4. How does a MOSFET work?

The voltage between Gate and Source (called V_{gs}) controls the switch. When V_{gs} is high enough, MOSFET turns ON; when V_{gs} is low, MOSFET turns OFF.

5. Why MOSFETs are popular

Compared to relays:

Advantages of using MOSFETs are:

- Fast switching
- Silent operation
- Long lifespan
- Low power consumption
- PWM capable

They also have their own limitations:

- Sensitive to wiring mistakes
- Can be damaged by voltage spikes
- Require proper gate drive

6. N-Channel vs P-Channel MOSFETs

N-Channel MOSFET

This is the most common type of MOSFET.

An N-Channel MOSFET has lower resistance, higher efficiency, and lower cost. It is mainly used in motor drivers, buck converters, and low side switching.

P-Channel MOSFET

P-channel MOSFET is less common compared to the N-Channel MOSFET.

While a P-Channel MOSFET rewards the user with easier high side switching, it has higher resistance and lower efficiency than an N-Channel one.

7. Logic Level MOSFETs

This is one of the most important concepts in MOSFETs. Not all MOSFETs can be controlled directly by a microcontroller.

Example: ESP32 output is 3.3V, but some MOSFETs require 10V on the gate for full operation.

Practical Rule

For microcontroller projects, use of logic-level MOSFETs is advised. Common examples are IRLZ44N, AO3400, IRLB8743.

8. Low-Side Switching

This is the most common beginner configuration.

In this configuration the MOSFET is placed between the load and ground.

Example:

Battery → Motor → MOSFET → Ground

Advantages

- Simple
- Efficient
- Easy to drive

This configuration is used in motors, LEDs, and solenoids.

9. High-Side Switching

MOSFET is placed between power supply and load.

This is usually more complex and requires additional circuitry.

Example:

Battery → MOSFET → Motor → Ground

10. Gate Resistor

Gate resistor is a small resistor placed between the microcontroller pin and MOSFET gate.

Its values usually lie between 100Ω – 330Ω

Adding this resistor limits switching current, reduces ringing, and protects GPIO pins.

11. Gate Pull-Down Resistor

A MOSFET gate tends to behave like a tiny capacitor.

Without a defined state, the gate might float. Leading to random unexpected switching.

Practical Rule

Add a $10k\Omega$ resistor between Gate and GND.

This ensures the MOSFET remains OFF when no signal is present.

12. Flyback Protection

Motors and relays are inductive loads.

When they are switched off, voltage spikes might occur.

A flyback diode is used to protect the MOSFET. This is one of the most common ways to protect your MOSFETs.

13. MOSFET Heating

Even when ON, a MOSFET is not a perfect conductor (since it is real and not ideal).

It has a resistance called $R_{DS(on)}$, which causes power loss.

Implying higher current means more heating.

Practical Implications

If a MOSFET becomes hot, check:

- Current draw
- Cooling
- MOSFET selection
- Wiring

Excessive heat reduces reliability.

14. PWM Control

MOSFETs are known for being able to switch rapidly, this allows them to perform Pulse Width Modulation (PWM).

PWM is used in motor speed control, LED dimming, and power control.

This is one of the biggest advantages of MOSFETs over relays.

Chapter Six: Grounding & Power Distribution

1. Why are Grounding and Power Distribution Important?

Every electronic circuit requires a source of power and a path for current to return. Many beginners think current only travels through the positive wire, but in reality current always flows in a complete loop.

Ground is the return path that completes this loop.

Poor grounding can cause:

- Random resets
- Unstable sensor readings
- Communication errors
- Noise problems
- Unexpected circuit behaviour

Good grounding improves stability, reliability, noise immunity, and makes troubleshooting easier.

2. What is Ground?

Ground is the common reference point of a circuit.

All voltages are measured relative to ground.

Example:

When we say $V_{ESP32} = 3.3V$ we actually mean,

$$V_{ESP32} - V_{ground} = 3.3V$$

Without a common ground reference, circuits cannot properly communicate.

3. Common Ground

Whenever two devices exchange signals, they must usually share a common ground.

Example: ESP32 controlling a motor driver

Required connections are signal wire and common ground.

Without common ground, the motor driver cannot correctly interpret the control signal.

Practical Rule

If two devices share signals, ensure that they share a ground reference.

This is one of the most common wiring mistakes.

4. Ground is Not Always Zero Volts

Ground is not magically the same everywhere.

Wires and PCB traces have resistance. So when current flows, voltage drops occur.

And as current increases, ground voltage can vary across the system. This is called Ground potential difference

5. Why Motors Cause Problems

Motors draw large amounts of currents.

When current flows through shared ground wiring, voltage drops appear.

This causes sensor errors, communication failures, and microcontroller resets.

Example:

Motor current and ESP32 current sharing the same thin ground wire.

Motor starts. Ground voltage shifts. ESP32 experiences an apparent voltage disturbance.

Reset occurs.

6. Power Distribution

Power should be distributed carefully throughout a system.

One of the common mistakes is connecting

Battery → Module 1 → Module 2 → Module 3

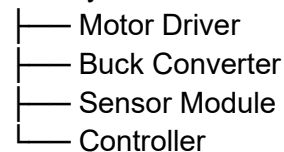
This causes voltage drops, shared current paths, increased noise.

Better Approach

Distribute power from a central point whenever possible.

Example:

Battery



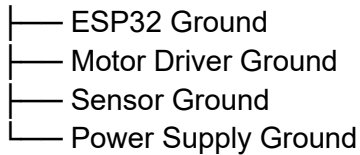
This reduces interaction between subsystems.

7. Star Grounding

Star grounding is a common technique in which all major grounds connect to a single central point.

Example:

Battery Ground



Star grounding reduces shared current paths, noise coupling, and makes troubleshooting easier.

8. Separating Power and Signal Grounds

High current devices like motors, servos, and switching regulators generate noise. Sensitive devices like IMUs, ADCs, and RF modules are affected by this generated noise.

Practical Rule

Keep noisy current paths away from sensitive circuits whenever possible.

Do not route motor currents through sensor grounds.

9. Ground Loops

A ground loop occurs when multiple return paths exist between two points.

Presence of ground loops can introduce noise, cause measurement errors and communication issues.

These parameters are particularly important in audio systems, measurement systems, and mixed analog/digital circuits.

For most robotics projects, the goal is to simply avoid unnecessary duplicate ground paths.

10. Decoupling and Grounding

Decoupling capacitors rely on good grounding.

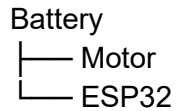
A decoupling capacitor works best when it is close to the IC, connected with short traces, and connected to a solid ground path.

Poor grounding reduces capacitor effectiveness.

11. Powering Motors and Logic

One of the most common robotics problems is a bad powering setup for motors and logic.

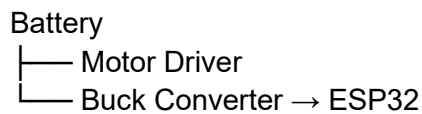
An example of a bad setup is:



with no consideration for current paths.

In such cases, motor startup causes voltage sag, noise, and resets.

A better setup would include:



Along with:

- A bulk capacitor near motor driver
- Decoupling capacitors near logic circuits.

This would significantly improve stability.

12. Grounding on PCBs

On PCBs, a ground plane is usually preferred.

It lowers the resistance and impedance, deals better with noise, and makes routing easier.

Ground planes improve both power distribution and signal integrity.

Chapter Seven: Noise & Filtering

1. Why are Noise & EMI Important?

In Practical electronic systems:

- Motors switch rapidly
- Regulators switch at high frequencies
- Long wires act like antennas
- High currents create voltage disturbances

These effects introduce unwanted electrical signals called noise.

Noise can cause:

- Random microcontroller resets
- Sensor inaccuracies
- Communication errors
- Unstable ADC readings
- Reduced system reliability

Understanding and controlling this noise is an important part of designing reliable systems.

2. What is Electrical Noise?

Any unwanted voltage or current variation present in a circuit is called electrical noise.

Ideally a 5V supply should remain at exactly 5V. But in reality small fluctuations occur due to switching, current spikes, and external interference.

These fluctuations are considered as noise.

3. What is EMI?

EMI(Electromagnetic Interference) is the unwanted electromagnetic energy that affects nearby circuits.

Motors, relays, servos, switching regulators, high current wiring, and wireless transmitters are common sources of EMI.

EMI can travel through wires, PCB traces, and air.

4. Common Sources of Noise

Motors

Motors are one of the largest noise sources.

They produce current spikes, voltage spikes, electromagnetic interference.

Due to these there can be ESP32 resets, sensor glitches, communication failures.

Switching Regulators

Buck converters rapidly switch current on and off, this creates ripple and high frequency noise.

While these are more efficient, they are noisier than linear regulators.

Long Wires

Long wires have resistance and inductance, which can pick up and radiate interference.

Long power and signal wires often become unexpected noise sources.

5. Symptoms of Noise Problems

Common symptoms are:

- Random resets
- ADC fluctuations
- I2C communication errors
- SPI communication instability
- Sensor spikes
- Wireless performance issues

Noise problems are often mistaken for software bugs.

6. Decoupling Capacitors

These are one of the most effective noise reduction methods. They provide a local source of energy for ICs.

When current demand changes suddenly, the capacitor supplies current before the power supply responds.

Practical Rule

Place a $0.1\mu F$ ceramic capacitor as close as possible to each IC's power pins.

This is one of the most important PCB design practices.

7. Bulk Capacitors

Bulk capacitors handle larger current demands.

They are commonly used for motor drivers, servos, and power rails.

Their typical values range from $100\mu F - 1000\mu F$.

Purpose

Bulk capacitors reduce voltage dips, startup surges, and sudden load changes.

8. Filtering

Filtering gets rid of unwanted noise while allowing useful signals or power to pass.

Capacitive Filtering

Capacitors naturally oppose rapid voltage changes.

They are used to reduce ripple, reduce switching noise, and stabilize supply rails.

Ferrite Beads

Ferrite beads act as high frequency resistors.

They pass DC current and suppress high frequency noise.

One can generally find them on power inputs, USB connections, and other sensitive circuits.

LC Filters

An LC filter combines Inductor and Capacitor.

It is used to reduce switching regulator ripple and power supply noise.

It can be found in audio circuits, RF systems, and precision electronics.

9. Noise Reduction Through Wiring

Good wiring practices significantly reduce noise.

Keep Wires Short

Long wires pick up interference and increase inductance.

Shorter wiring generally improves performance.

Twist Power Wires Together

Twisting power & ground and motor supply wires helps reduce electromagnetic emissions and susceptibility to noise.

Separate Power & Signal Wiring

Avoid routing motor and servo wires alongside sensor and communication lines.
Physical separation reduces interference.

10. Noise Reduction Through Grounding

Grounding plays a major role in noise control.

Poor grounding can spread noise and create voltage offsets; while good grounding improves stability, signal quality, and power integrity.

Refer to Chapter 6 for grounding practices.

11. Noise Reduction Through Layout

PCB layout also strongly influences performance.

Practical rules:

- Use a ground plane
- Keep current loops small
- Place decoupling capacitors close to ICs
- Keep switching regulators away from sensitive circuits.

Good layout often solves noise problems before they occur.

Chapter Eight: Battery & Protection Design

1. Why is Battery & Protection Design important?

Many embedded and robotics systems rely on batteries as their primary power source.

Unlike wall adapters:

- Battery voltage changes over time
- Batteries have current limits
- Batteries can be damaged by misuse
- Batteries can become safety hazards if improperly handled

Good battery design improves reliability, runtime, safety, and system lifespan.

2. Common Battery Types

Lithium-Ion (Li-ion)

18650 and 21700 cells are common examples of Li-ion batteries.

Li-ion batteries generally have high energy density, and are rechargeable.

Nominal Voltage: 3.7V

Fully Charged Voltage: 4.2V

Recommended Minimum: 3.0V

Common applications of Li-ion batteries are power banks, robots, and portable electronics.

Lithium Polymer (LiPo)

LiPo batteries are very similar to Li-ion, but they differ in some aspects such as:

- They come in pouch construction
- Are more lightweight
- And available in higher discharge ratings

They are common in drones, RC vehicles, and high performance robotics.

Nickel Metal Hydride (NiMH)

NiMH is an older rechargeable technology.

Although it is safer to use and more tolerant of abuse, it comes with a lower energy density.

It is commonly found in AA rechargeable batteries and educational projects.

3. Understanding Battery Voltage

Battery voltage is not constant.

Designs must accommodate the entire voltage range

One of the most common beginner mistakes is designing for nominal voltage only.

4. Series and Parallel Connections

Series Connection

Voltages add. Capacity remains unchanged

Example: Two 3.7V cells in series give 7.4V output but their Ah rating remains the same.

Parallel Connection

Capacity adds. Voltage remains unchanged.

Example: Two 2000mAh cells in parallel give a 4000mAh output but their output voltage rating is the same as a single cell.

5. Battery Capacity

Capacity describes how much charge a battery can store.

Units:

- Ah – Ampere hour
- mAh – milliampere hour

1000mAh = 1Ah

Higher capacity generally means longer runtime and larger battery size

6. Estimating Runtime

Approximate runtime of a battery can be calculated using the following relation:

$$\text{Runtime} = \frac{\text{Battery Capacity}}{\text{Load Current}}$$

Example:

If a 500mA load is applied on a 2000mAh battery, its runtime is approximated to 4 hours.

We consider this calculated runtime as an approximation as actual runtime is usually lower due to efficiency losses and varying loads.

7. Understanding C-Rating

C-rating describes how quickly a battery can safely deliver current.

Example:

If a 2000mAh battery has a 10C rating

Then the maximum current it can deliver is

$$2A \times 10 = 20A$$

Higher C-ratings allow larger current delivery.

This is particularly important for motors, drones, and mobile robots.

8. Battery Management System (BMS)

A BMS protects rechargeable battery packs.

Functions of a good BMS typically include:

- Overcharge protection
- Over-discharge protection
- Overcurrent protection
- Short-circuit protection

A BMS improves both safety and battery lifespan.

9. Over-Discharge Protection

Lithium batteries should not be deeply discharged as excessive discharge can reduce capacity and permanently damage cells.

Protection circuits or BMS boards are commonly used to prevent this.

10. Overcharge Protection

Lithium batteries should also not be charged above their specific voltage.

Overcharging can cause cell damage, excessive heating, and other safety hazards.

Hence, proper charging circuits are essential.

11. Reverse Polarity Protection

Connecting a battery backwards can destroy electronics instantly.

Protection methods include diodes or MOSFET based protection circuits.
Reverse polarity protection is strongly recommended for portable systems.

12. Fuses

A fuse protects against excessive current.

When current exceeds its rating, the fuse disconnects the circuit.

Fuses help protect wiring, batteries, and electronics.

Practical Rule

A fuse should be placed as close to the battery as practically possible.
This minimizes the length of unprotected wiring.

13. Power Switches

A proper power switch improves safety and usability.
Switches should be capable of handling system voltage and maximum expected current.

Undersized switches may overheat or fail.

14. Battery Selection guidelines

Voltage Requirements

Can the battery provide the required voltage range?

Current Requirements

Can the battery safely deliver peak current?

Runtime Requirements

Does the capacity provide sufficient operating time?

Size and Weight Constraints

Particularly important in mobile systems.

Chapter Nine: PCB Power Layout Rules

1. Why are PCB Power Layout Rules important?

A circuit can be electrically correct and can still fail due to bad or poor PCB layout.

In practical circuits high currents cause voltage drops, switching circuits generate noise, long traces act as inductors, poor grounding creates instability.

Poor layout can result in random mcu resets, excessive noise, voltage ripple, overheating of traces, reduced efficiency, and difficult debugging.

A good PCB layout improves reliability, efficiency, noise stability, thermal performance

2. Trace Width

PCB traces have resistance.

As current increases: Voltage drop increases, and heat generation increases.

Hence, traces carrying high current can become hot and cause performance issues.

Practical Rule

Use Wider traces for:

- Motor power
- Battery Connections
- Regulator Outputs
- High current loads

Signal traces can be relatively thin.

Common Mistakes

Use the same trace width for ESP32 GPIO and motor supply.

Power traces should always be given special consideration.

3. Keep High Current Paths Short

Every trace has resistance and inductance.

Long high current paths lead to increased voltage drops, increased noise, and heating.

Practical Rule

Keep paths short between:

- Battery and motor driver
- Regulator and load
- Power connector and PCB

Current must take the shortest practical path.

4. Decoupling Capacitor Placement

Decoupling capacitors only work effectively when placed close to the device they protect.

Long traces reduce their effectiveness.

Practical Rule

Place a ceramic decoupling capacitor:

- As close as possible to the IC power pins
- Using short, direct connections

Typical values are $0.1\mu F$ and $1\mu F$ ceramic capacitors.

Common Mistakes

Placing all capacitors together in one corner of the PCB.

5. Bulk Capacitor Placement

Bulk capacitors provide energy during sudden current demands or spikes.

Examples:

- Motor startup
- Servo movement
- Radio transmission bursts

Practical Rule

Place bulk capacitors close to:

- Voltage regulators
- Motor drivers
- High-current modules

Typical values range from $100\mu F$ to $1000\mu F$ electrolytic.

6. Ground Planes

Ground is not “everywhere the same”.

Current must return to its source.

A poor grounding route can cause noise, ground bounce, and communication issues.

Practical Rule

Use a ground plane whenever possible

Benefits:

- Lower resistance
- Lower impedance
- Better noise performance
- Easier routing

Ground planes are generally preferred over thin ground traces.

7. Separate Noisy & Sensitive Circuits

Elements of some circuits generate noise. Common examples include motors, motor drivers, and switching regulators.

But some circuits are sensitive to the noise generated by these components, such as MCUs, ADCs, sensors, and RF modules.

Practical Rule

Keep noisy circuits physically separate from noise sensitive circuits.

Example:

Place the motor driver section away from the IMU and ESP32 antenna.

8. Minimize Current Loops

Current always flows in a loop, and large loops can act as antennas and radiate noise.

Practical Rule

Keep supply and return paths close together.

Example:

- Route power and ground together
- Keep switching regulator loops compact

Smaller loop area = Less EMI

9. Connector Placement

Connectors are one of the most overlooked components in PCB layout considerations.

A poor placement of connectors can cause long power paths, voltage drop, and difficult routing.

Practical Rule

Place connectors close to the circuitry they serve.

Example:

- Battery connector near power input stage
- Motor connector near motor driver.

10. Thermal Considerations

Copper traces help dissipate heat.

Hence, larger copper areas help reduce temperature rise and improve reliability.

Practical Rule

Provide additional copper around voltage regulators, MOSFETs, and motor drivers.

High power components must not be isolated on tiny copper islands.

Chapter Ten: Reliability & Thermal Considerations

1. Why are Reliability & Thermal considerations important?

Every electronic component has operating limits.

In practical systems regulators dissipate heat, MOSFETs generate heat during switching, motors draw large current, and high current traces experience losses.

Excessive heat can lead to reduced efficiency, unstable operation, shortened component lifespan, and sometimes permanent component failure.

A system that works for 5 minutes but overheats after 30 minutes is not a reliable system.

Reliability is the ability of a system to operate correctly over long periods of time under expected operating conditions.

2. Why components heat up

Heat is produced whenever electrical energy is lost.

Common sources of heat include resistive losses, voltage regulator losses, and motor driver losses.

Resistive Losses

Whenever current flows through a resistance, the power lost is described by $P = I^2 R$.

This applies to wires, PCB traces, connectors, MOSFETs, motors.

As current increases, heat increases rapidly.

Voltage Regulator Losses

Linear regulators dissipate excess voltage as heat

$$P = (V_{in} - V_{out}) I$$

Large voltage drops and high current result in significant heating.

This is one of the most common causes of overheating in projects.

Motor Driver Losses

Motor drivers handle large currents and heat in this case is generated due to internal transistor resistance, switching losses, and stall current conditions.

Motor drivers often become the hottest component in a robot.

3. Why heat is a problem

Heat affects more than just comfort.

Excessive rise in temperature can cause unexpected MCU resets, sensor drift, reduced battery performance, component degradation, and sometimes permanent failure.

Many failures that occur randomly are usually thermal problems.

4. Temperature and component Lifespan

Higher operating temperatures generally reduce component lifespan.

Example:

- Electrolytic capacitors age faster
- Batteries degrade faster

Keeping components cool improves long term reliability

5. Heatsinks

A heatsink increases surface area and improves heat dissipation.

It is commonly used with linear regulators, motor drivers, and power MOSFETs.

Practical Rule

If a component is too hot to comfortably touch after normal operation, thermal management should be investigated.

6. Airflow

Heat must be removed from the system.

Even a small amount of airflow can significantly reduce temperatures.

Practical Rule

Avoid enclosing heat generating components in completely sealed spaces without ventilation.

Special attention should be paid to battery compartments, robot chassis, power supply enclosures.

7. Component derating

Components should not be operated continuously at their maximum ratings.

This makes them prone to reduced lifespan.

Practical Rule

Use components at approximately 70-80% of their rated limits.

Example:

- A 10A MOSFET should not be expected to continuously carry 10A in a real system
- A 3A regulator is more reliable when supplying 2A rather than 3A continuously.

This provides safety margin and improves lifespan.

8. Reliability through margin

Good designs include margin for current, voltage, and temperature.

Examples:

- Use a 25V capacitor on a 12V rail rather than 16V capacitor
- Use a regulator rated for more current than the expected load.
- Avoid designing systems that operate at their absolute limits.

Chapter Eleven: System Level Power Design Checklist

1. Before Connecting Power

Power Source Verification

- ☐ Supply voltage is correct
- ☐ Polarity is correct
- ☐ Supply current capability is sufficient
- ☐ Battery voltage range has been verified
- ☐ Protection circuitry is present where required

Component Ratings

- ☐ Capacitor voltage ratings are sufficient
- ☐ Diode ratings are sufficient
- ☐ MOSFET ratings are sufficient
- ☐ Regulator ratings are sufficient
- ☐ Connector ratings are sufficient

2. Power Distribution Checks

- ☐ Common ground exists between communicating devices
- ☐ Ground wiring is adequately sized
- ☐ High-current and low-current paths are separated
- ☐ Power is distributed from a central point where practical
- ☐ Return current paths have been considered

3. Capacitor Checks

- ☐ Decoupling capacitors are present near ICs
- ☐ Bulk capacitors are present near high-current loads
- ☐ Capacitors are correctly polarized
- ☐ Capacitors are placed close to the circuits they support

4. Regulator Checks

- ☐ Input voltage is within regulator limits
- ☐ Output voltage has been verified
- ☐ Regulator current rating is sufficient

- ☐ Heat dissipation has been considered
- ☐ Required input and output capacitors are present

5. Diode Checks

- ☐ Diode orientation is correct
- ☐ Flyback protection is present for inductive loads
- ☐ Reverse polarity protection is present where required
- ☐ Voltage drop has been considered

6. MOSFET Checks

- ☐ Appropriate MOSFET selected
- ☐ Logic-level MOSFET used when required
- ☐ Gate resistor installed
- ☐ Gate pull-down resistor installed
- ☐ Flyback protection present for inductive loads
- ☐ Expected current within safe limits

7. Battery Checks

- ☐ Battery voltage range understood
- ☐ Battery capacity meets runtime requirements
- ☐ Battery can supply required peak current
- ☐ BMS present when required
- ☐ Fuse installed near battery
- ☐ Charging method is appropriate

8. Noise & EMI Checks

- ☐ Decoupling capacitors installed
- ☐ Bulk capacitors installed
- ☐ Noisy circuits separated from sensitive circuits
- ☐ Wiring kept reasonably short
- ☐ Motor wiring separated from signal wiring
- ☐ Grounding practices followed

9. PCB Layout Checks

- ☐ High-current traces are sufficiently wide
- ☐ Ground plane present
- ☐ Decoupling capacitors close to ICs
- ☐ Current loops minimized

- ☐ High-current paths kept short
- ☐ Thermal dissipation considered

10. Thermal & Reliability Checks

- ☐ No component operates continuously at maximum rating
- ☐ Regulators remain within safe temperatures
- ☐ MOSFET temperatures are acceptable
- ☐ Adequate airflow available
- ☐ Safety margin included in design

11. Before Connecting Motors

- ☐ Motor supply verified
- ☐ Driver ratings verified
- ☐ Flyback protection present
- ☐ Bulk capacitance installed
- ☐ Wiring sized appropriately
- ☐ Logic and motor systems properly isolated

12. First Power On Procedure

Step 1

Power the system without the load connected. Then verify:

- ☐ Correct voltages
- ☐ No overheating
- ☐ No unexpected current draw

Step 2

Connect low-power loads and verify:

- ☐ Stable operation
- ☐ Stable communication
- ☐ No excessive heating

Step 3

Connect high-current loads and verify:

- ☐ Stable voltage
- ☐ No resets
- ☐ No excessive noise
- ☐ Expected current consumption

13. Troubleshooting Quick Guide

Random Resets

Check:

- ☐ Battery voltage sag
- ☐ Regulator capacity
- ☐ Bulk capacitors
- ☐ Grounding
- ☐ Motor noise

Unstable Sensor Readings

Check:

- ☐ Grounding
- ☐ Decoupling capacitors
- ☐ Wiring layout
- ☐ Noise sources

Excessive Heating

Check:

- ☐ Current draw
- ☐ Voltage drop
- ☐ Regulator losses
- ☐ MOSFET selection
- ☐ Airflow

Motors Behaving Erratically

Check:

- ☐ Power supply capability
- ☐ Flyback protection
- ☐ Wiring
- ☐ Driver ratings
- ☐ Grounding

14. Final Design Review

Before considering a design complete:

- ☐ Electrical ratings verified
- ☐ Power distribution verified
- ☐ Protection systems verified
- ☐ Thermal performance verified
- ☐ Noise mitigation verified
- ☐ Battery system verified
- ☐ Long-duration testing completed
- ☐ System behavior understood under worst-case conditions